

Assignment:3

Objective

The objective of this assignment is to analyze the Superstore sales dataset using advanced SQL concepts such as Subqueries, Common Table Expressions (CTEs), and Window Functions. The dataset is organized into separate tables to improve data management and query efficiency. Various analytical queries are then performed to gain meaningful business insights from customer and sales data.

Step 1: Data Preparation and Table Creation

The Superstore dataset is initially imported into a raw table named **superstore_raw**. To achieve better database normalization and organization, the data is divided into three separate tables:

- **Customers Table** – Stores customer-related information such as customer ID, customer name, and segment.
- **Products Table** – Contains product details including product ID, product name, category, and sub-category.
- **Orders Table** – Stores order-specific information such as order dates, shipping details, sales, quantity, discount, and profit.

Distinct records are extracted from the raw dataset to ensure data consistency and eliminate duplicate entries.

Step 2: Query Analysis

1. Orders with Sales Greater Than Average

A subquery is used to calculate the average sales value across all orders. The main query then identified orders whose sales exceeded this average. This analysis helps highlight high-value transactions and provides insight into above-average sales performance.

46 FROM orders
47)

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Product ID	Sales	Quantity	Discount	Profit
CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0	41.9136
CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2	90.7152
CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002033	911.424	4	0.2	68.3568
CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	OFF-BI-10003656	407.976	3	0.2	132.5922
CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0	13.3176
CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0	240.2649
US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5	-1665.0522
CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2	123.4737
CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32	-46.9764
CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	TEC-PH-10000486	371.168	4	0.2	41.7564
CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	TEC-PH-10002447	1029.95	5	0	298.6855
CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	FUR-CH-10003968	319.41	5	0.1	7.098
CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	OFF-AR-10002671	1113.024	8	0.2	111.3024

2. Highest Sales Order for Each Customer

A correlated subquery is utilized to determine the maximum sales amount for every customer. The query returned the highest-value order placed by each customer, making it easier to identify significant purchasing behavior and top transactions.

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Product ID	Sales	Quantity	Discount	Profit
CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	219.582
US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	-383.031
CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	85.3092
CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0	240.2649
US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5	-1665.0522
CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2	123.4737
CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32	-46.9764
CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	TEC-PH-10002447	1029.95	5	0	298.6855
CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	FUR-CH-10003968	319.41	5	0.1	7.098
CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	OFF-AR-10002671	1113.024	8	0.2	111.3024
CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	TEC-AC-10000844	339.96	5	0.2	67.992
CA-2015-110744	9/7/2015	9/12/2015	Standard Class	HA-14920	OFF-ST-10003656	671.93	7	0	20.1579
CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	FUR-CH-10000454	1951.84	8	0	585.552
CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	TEC-MA-10000822	8159.952	8	0.4	-1359.992
CA-2014-118962	8/5/2014	8/9/2014	Standard Class	CS-12130	FUR-CH-10003817	340.144	7	0.2	21.259
CA-2014-165101	12/5/2014	12/10/2014	Standard Class	DK-13150	TEC-AC-10000450	408.744	7	0.2	76.6305

orders 16 x

Output

3. Total Sales for Each Customer

A Common Table Expression (CTE) is used to calculate the cumulative sales generated by each customer. This provided a clear overview of customer contribution to overall business revenue and simplified further analysis.

Result Grid			Filter Rows:
	Customer ID	Total_Sales	
▶	CG-12520	1148.7800000000002	
	DV-13045	1119.4830000000002	
	SO-20335	2602.5755000000001	
	BH-11710	6255.3510000000001	
	AA-10480	1782.872	
	IM-15070	4930.4739999999999	
	HP-14815	1990.314	
	PK-19075	8158.654	
	AG-10270	2565.25799999999994	
	ZD-21925	1493.944	
	KB-16585	2742.75800000000003	
	SF-20065	763.54599999999998	
	EB-13870	2767.219	
	EH-13945	684.171	684.
	TB-21520	4730.628	
	MA-17560	4280.6209999999999	
Result 17			×

4. Customers with Above-Average Total Sales

Using a combination of a CTE and a subquery, customers whose total sales exceeded the average customer sales are identified. This helped distinguish high-performing customers who contribute more revenue than the average customer base.

Result Grid			Filter Rows:
	Customer ID	Total_Sales	
▶	BH-11710	6255.3510000000001	
	IM-15070	4930.4739999999999	
	PK-19075	8158.654	
	TB-21520	4730.628	
	MA-17560	4280.6209999999999	
	SN-20710	3127.95920000000007	
	LC-16930	4481.162	
	RA-19885	3832.31399999999994	
	ES-14080	4648.516	
	KM-16720	4909.4720000000001	
	KD-16270	8282.358	
	JE-15745	6400.4549999999999	
	SC-20770	4492.6569999999999	
	CS-12400	3651.85850000000007	
	GM-14455	3078.62400000000003	
	RB-19465	4556.5660000000002	
Result 18			×

5. Ranking Customers Based on Total Sales

A window function is applied to rank customers according to their total sales. Customers generating higher revenue received higher rankings. This ranking mechanism is useful for customer segmentation and business decision-making.

	Customer ID	Total_Sales	Sales_Ran
▶	SM-20320	25043.05	1
	TC-20980	19017.847999999998	2
	RB-19360	15117.339	3
	TA-21385	14595.62	4
	AB-10105	14355.610999999997	5
	SC-20095	14142.333999999999	6
	KL-16645	14071.917	7
	HL-15040	12873.297999999999	8
	SE-20110	12209.438000000002	9
	CC-12370	12129.072	12209.43800000
	TS-21370	11885.871	11
	GT-14710	11820.119999999997	12
	BM-11140	11609.900000000001	13
	SV-20365	11420.645999999997	14
	CJ-12010	11079.742	15
	CL-12565	10880.545999999998	16

Result 19 ×

6. Row Number Assignment Within Each Customer

The ROW_NUMBER() window function is used with PARTITION BY to assign sequential numbers to orders belonging to the same customer. This made it possible to analyze customer order history and identify the order sequence for each customer.

	Order ID	Customer ID	Sales	Row_Num
▶	CA-2016-103982	AA-10315	3930.072	1
	CA-2014-128055	AA-10315	673.568	2
	CA-2016-103982	AA-10315	431.976	3
	CA-2017-147039	AA-10315	362.94	4
	CA-2014-128055	AA-10315	52.98	5
	CA-2016-103982	AA-10315	41.72	6
	CA-2015-121391	AA-10315	26.96	7
	CA-2014-138100	AA-10315	14.94	8
	CA-2014-138100	AA-10315	14.56	9
	CA-2017-147039	AA-10315	11.54	10
	CA-2016-103982	AA-10315	2.304	11
	CA-2016-131065	AA-10375	499.98	1
	CA-2015-140921	AA-10375	149.97	2
	CA-2017-100230	AA-10375	116.98	3
	CA-2015-114503	AA-10375	84.96	4
	US-2017-169488	AA-10375	39.96	5

Result 20 ×

7. Top Three Customers by Total Sales

Customers are ranked according to their total sales, and the top three performers are selected. This analysis helps identify the most valuable customers and supports targeted customer relationship strategies.

	Customer ID	Total_Sales	Sales_Rank
▶	SM-20320	25043.05	1
	TC-20980	19017.847999999998	2
	RB-19360	15117.339	3

Step 3: Final Combined Analysis

The final query combined multiple SQL concepts, including JOIN operations, CTEs, and Window Functions. The query displayed:

- Customer Name
- Total Sales
- Sales Rank

The CTE first calculated total sales for each customer. The results are then joined with the customer information table to retrieve customer names. Finally, a ranking function is applied to rank customers based on their overall sales contribution.

This combined analysis provides a comprehensive overview of customer performance and enables businesses to quickly identify their most valuable customers.

Result Grid |  Filter Rows: | Export:  | W

	Customer Name	Total_Sales	Sales_Rank
▶	Sean Miller	25043.05	1
	Tamara Chand	19017.847999999998	2
	Raymond Buch	15117.339	3
	Tom Ashbrook	14595.62	4
	Adrian Barton	14355.610999999997	5
	Sanjit Chand	14142.333999999999	6
	Ken Lonsdale	14071.917	7
	Hunter Lopez	12873.297999999999	8
	Sanjit Engle	12209.438000000002	9
	Christopher Conant	12129.072	10
	Todd Sumrall	11885.871	11
	Greg Tran	11820.119999999997	12
	Becky Martin	11609.900000000001	13
	Seth Vernon	11420.645999999997	14
	Caroline Jumper	11079.742	15
	Clay Ludtke	10880.545999999998	16
	Maria Etezadi	10635.917999999998	17

Result 23 ×

Output